

TRACKER SERVICE CHANNEL



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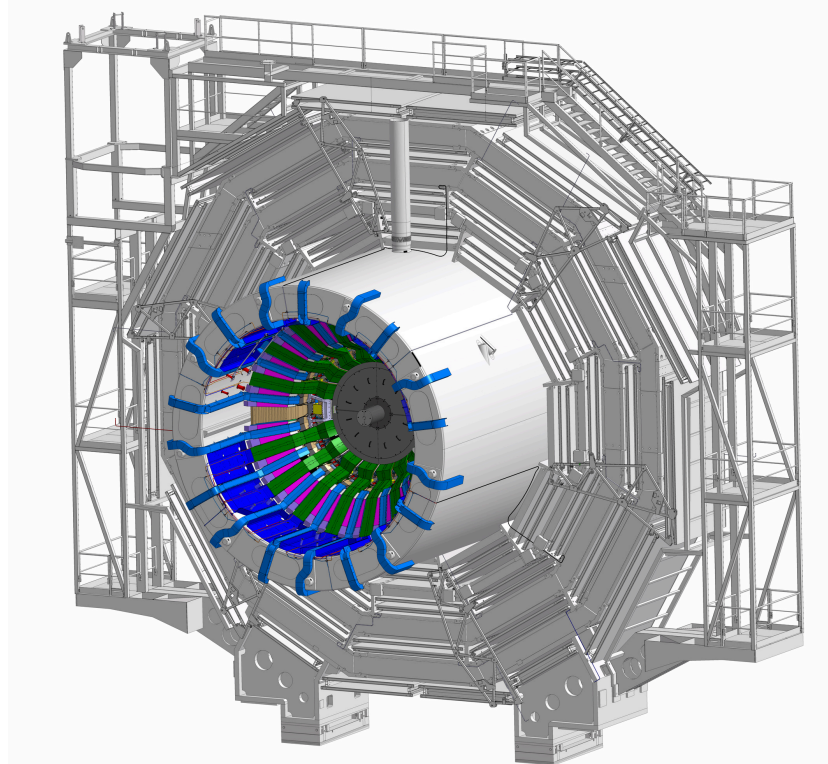
Tracker Service Channel Context

As part of the CMS Phase-2 upgrade, the CMS Tracker cooling system will be upgraded to operate at temperatures down to -35°C . Liquid CO_2 will be distributed through: Detector Transfer Lines (DTLs); Patch Panel 1 (PPI); Tracker Service Channels (TSCs) and finally reaching the detector modules inside the Tracker.

Due to the limited space available inside the TSCs, their design requires careful optimisation. In particular, condensation forming on the outer surface of the service channels could compromise detector operation and system reliability.

This poster presents:

- Results of extensive thermal simulations performed on a new TSC design during the Phase-2 preparation phase
- Thermal test results obtained from a full-scale (1:1) prototype



Purpose of the Service Channels

The main purpose of the tracker service channels is to safely guide tracker services inside the CMS detector.

Tracker services connect individual detector modules with their terminations. They include:

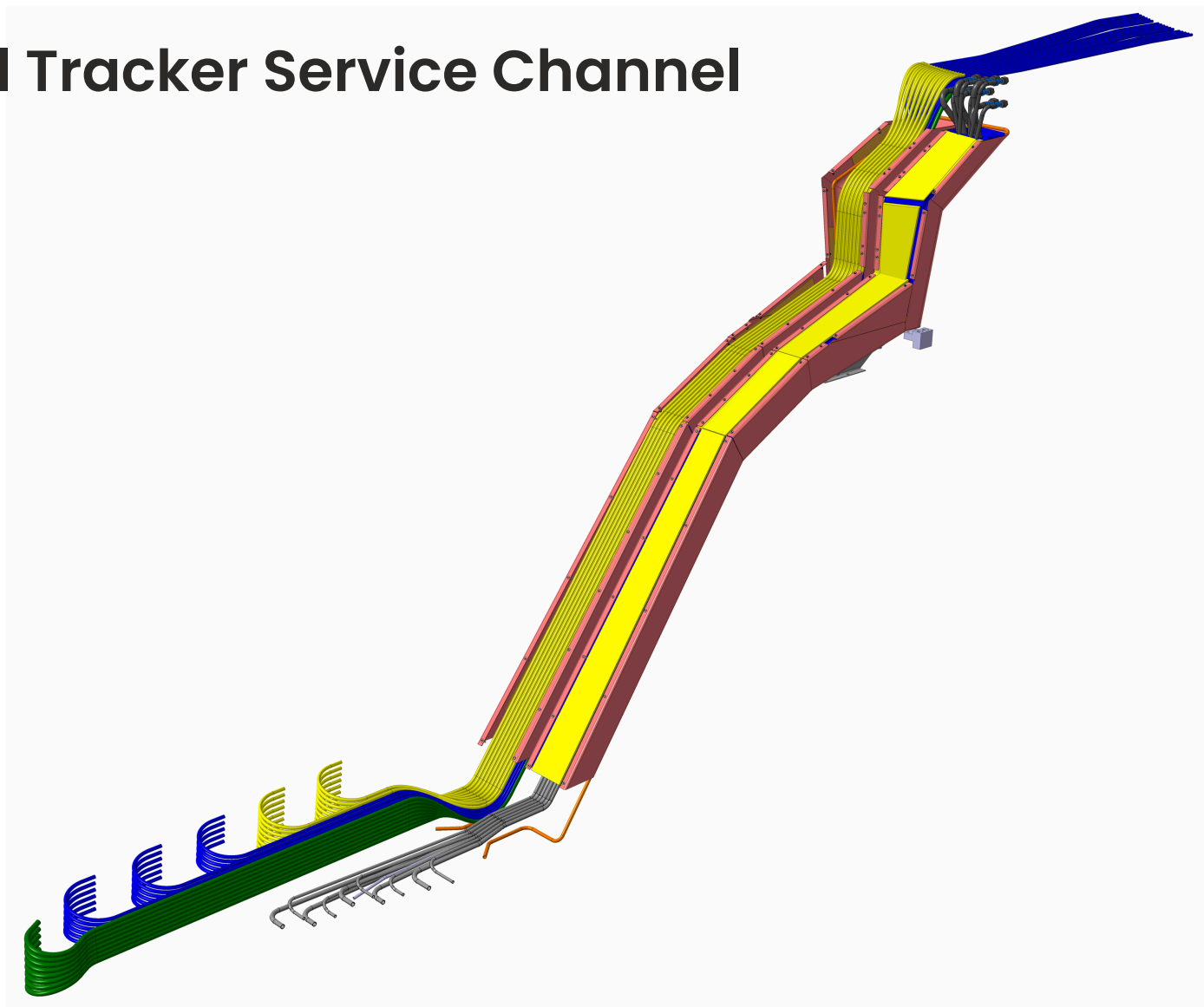
- Electrical cabling for module powering; Optical fibres for data transfer; A CO_2 cooling circuit to maintain the detectors at their ideal operating temperature

System overview

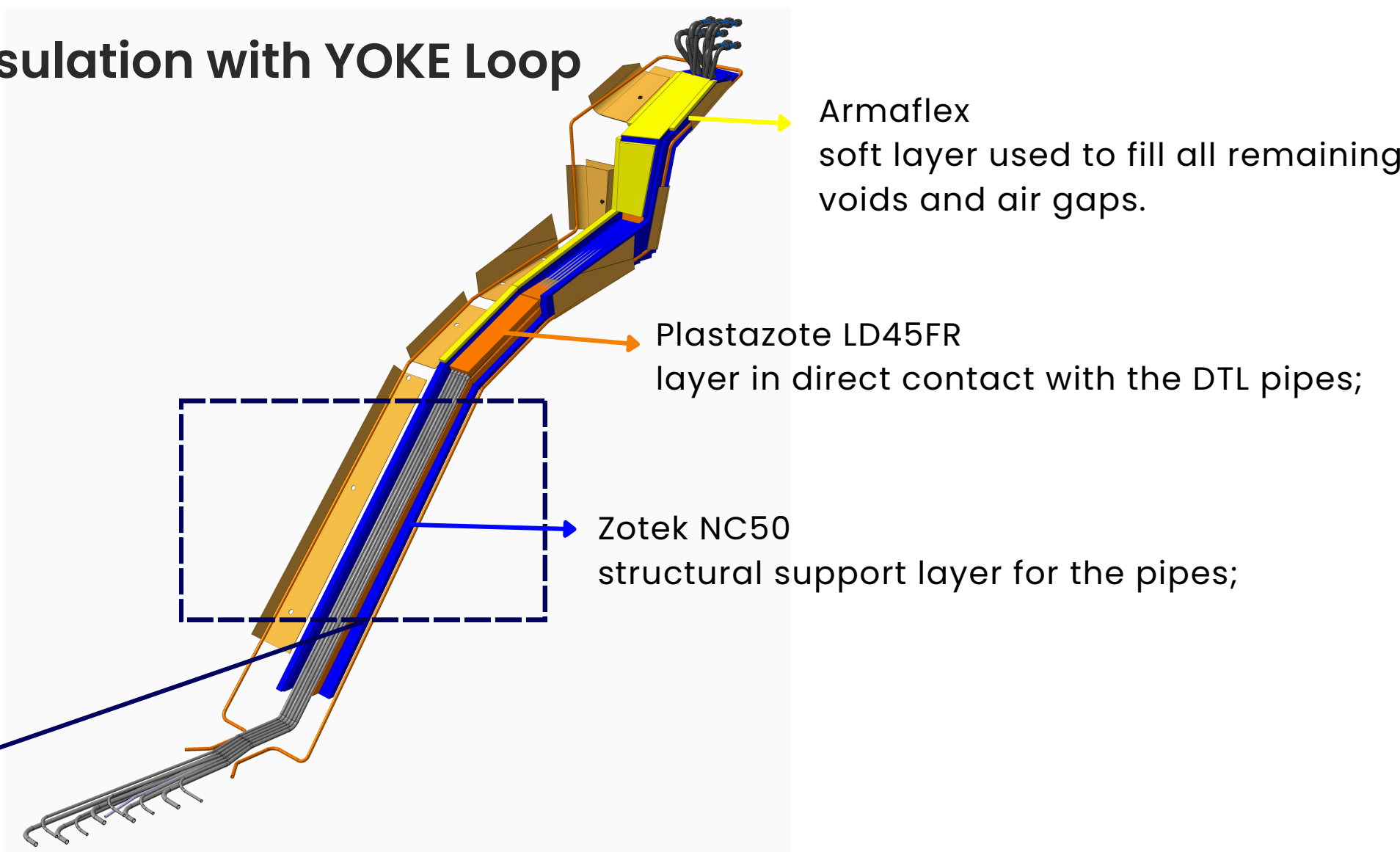
- Total number of service channels: 32
 - Different configuration variants: multiple
 - Cooling variant reproduced: 20 times
 - Total insulation volume: $\sim 0.65\text{ m}^3$
- Each tracker service channel is composed of 5 separate sections.

Tracker Service Channel Design

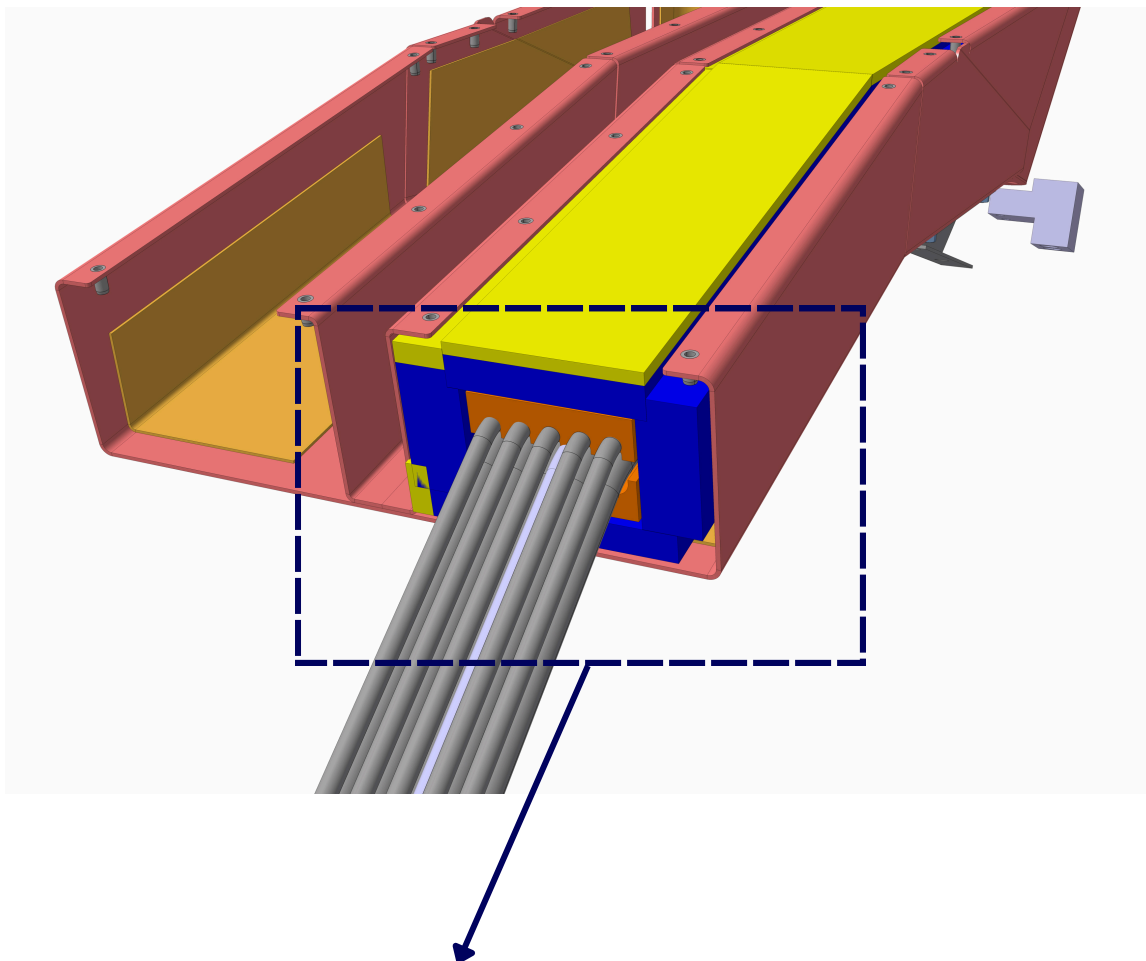
Full Tracker Service Channel



Insulation with YOKE Loop



Cross Section of the TSC



The water loop is made from a 12 mm OD copper pipe soldered to copper plates, which are mechanically attached to the aluminium TSC sections. The copper plates improve heat distribution and ensure efficient heat transfer to and from the circulating water.

The inlet is located at the cooling section, while the outlet is positioned at the sections carrying the power supply cables.

The purpose of the water loop is twofold:

- to maintain the temperature of the TSC above the dew point in the cold section, preventing condensation;
- to cool the power supply cables, which can reach temperatures of approximately 40°C .



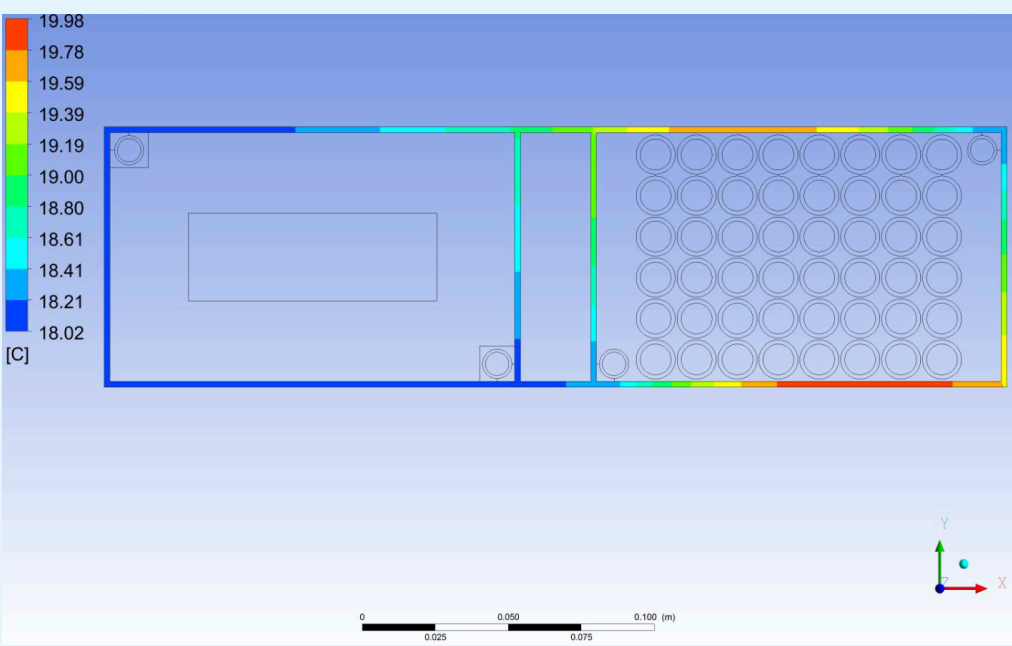
Carefully designed insulation pieces eliminate cold bridges and air pockets around the DTL cold lines, significantly reducing the risk of condensation on the TSC.



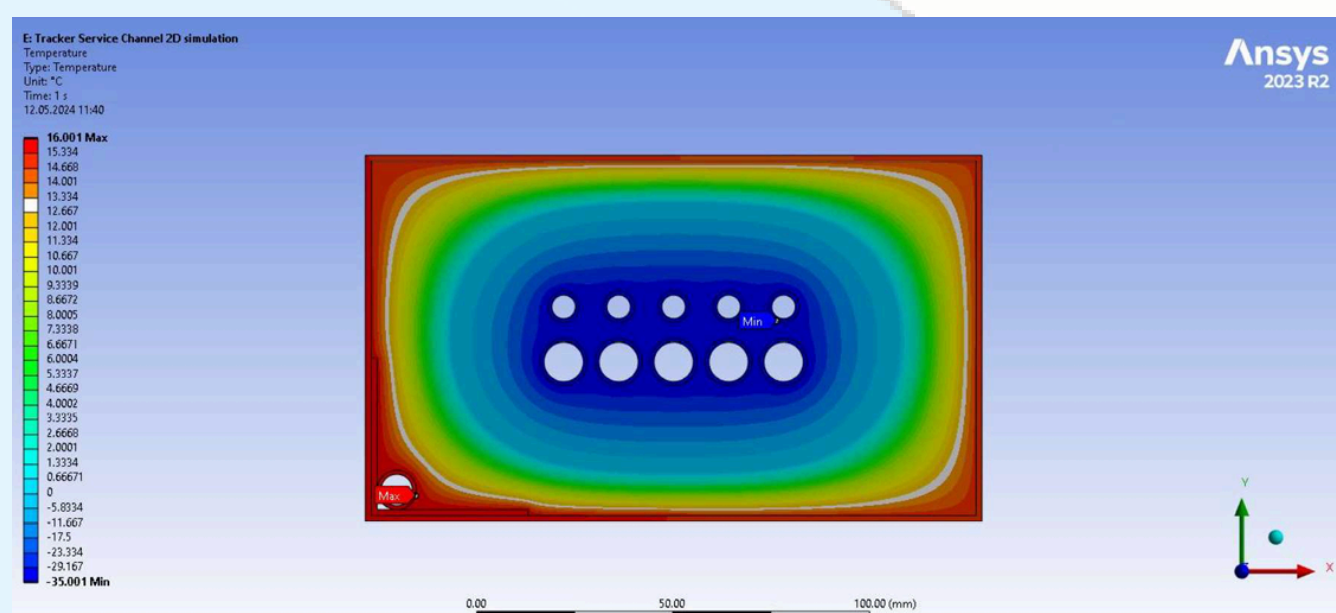
Simulations

The simulations were conducted to optimize the TSC design. Their purpose was to evaluate how the TSC material affects surface temperatures and to assess whether the selected insulation materials provide a sufficient thermal barrier for the system.

The aluminium frame shows higher overall temperatures compared to the stainless steel structure, indicating more effective heat distribution. This improved thermal conductivity reduces the risk of localized cold spots.



TSC made in aluminium.



Temperatures over the TSC cross-section

Thermal Test

The thermal performance was evaluated using a full-scale Tracker Service Channel.

The prototype included a water cooling loop with copper heat-spreading plates and was fully equipped with the final insulation configuration and material. To minimize the influence of air convection, the setup was surrounded by insulation panels, while the ambient conditions were kept close to those expected inside the cavern.

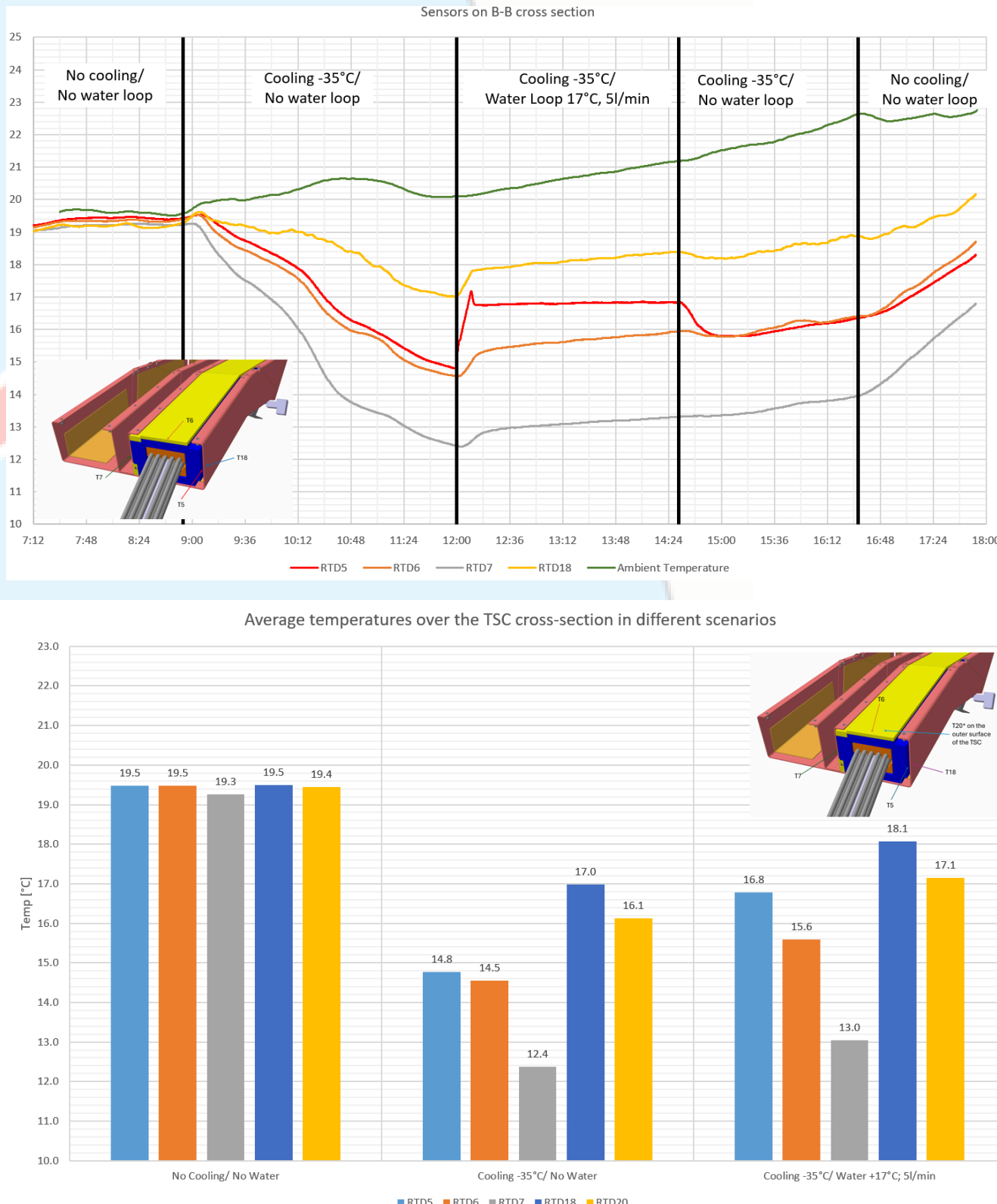
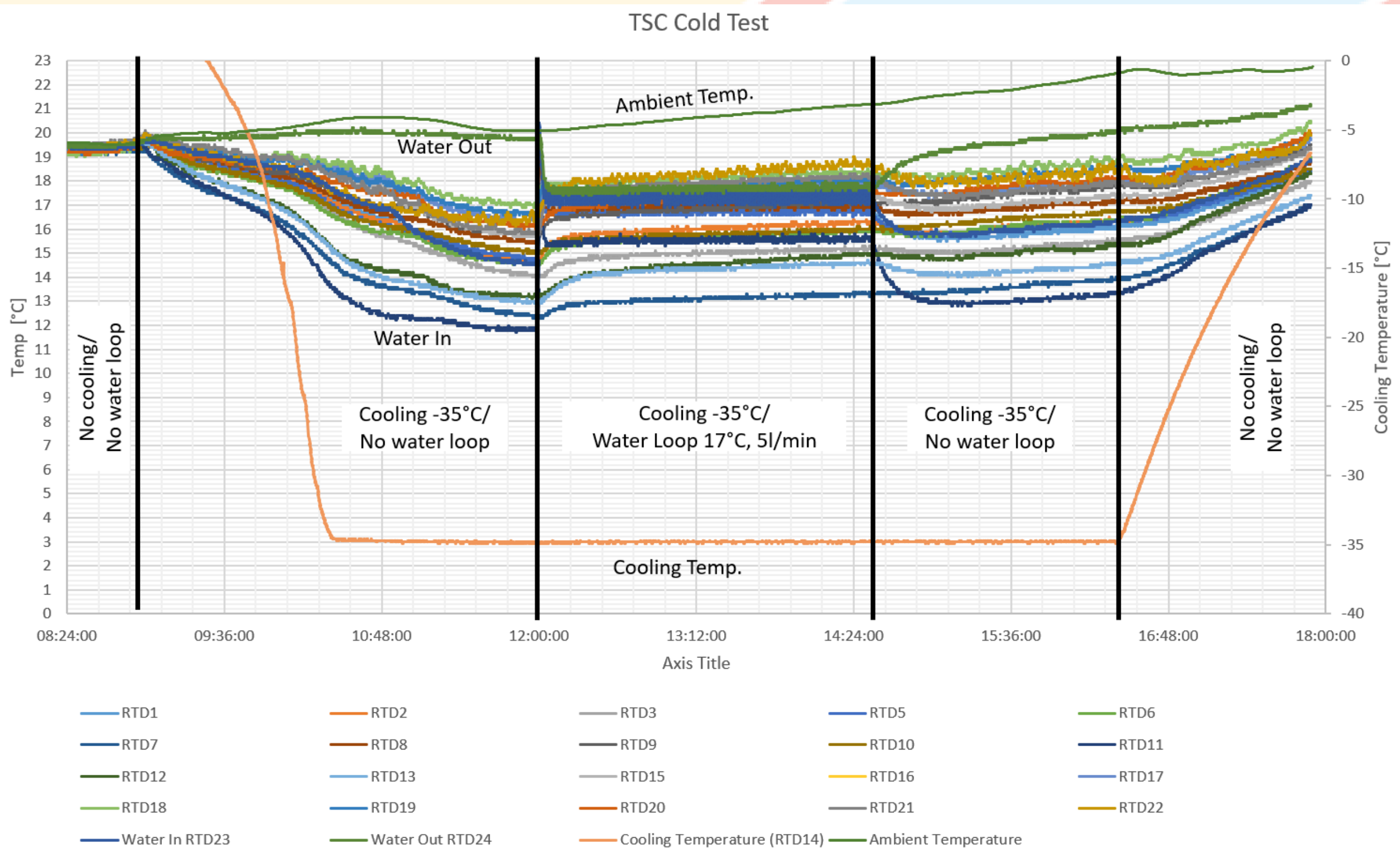
During the test:

- The DTL pipe temperature was set to -35°C
- The water loop operated at 17°C with a flow rate of 5 L/min
- Cooling of the dry gas flowing through the OD 6 mm copper pipe was also monitored
- The inlet gas temperature matched the ambient temperature (20°C) with a flow rate of 250 L/h.

Results

The measurements show that the insulation alone is sufficient to maintain the TSC surface temperature above the dew point. Thermal imaging confirmed that the aluminium sections distribute heat efficiently across the structure, preventing the formation of localized cold spots on the outer surface.

The dry gas temperature decreased by approximately 20 K, reaching -2°C at the outlet.



Tracker Service Channel- setup

